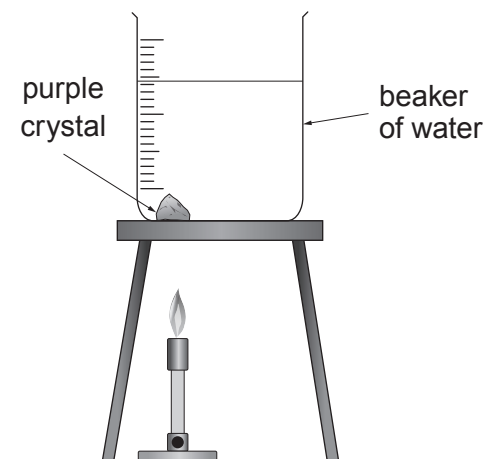


3. A group of students investigated different methods of heat transfer.

(a) First, they added a purple crystal to a beaker of water and heated it up as shown below.



(i) **Circle** the method of heat transfer that was being investigated.

conduction

convection

radiation

[1]

(ii) I. State what the students observed. You may add to the diagram if you wish.

[1]

.....

.....

II. Give a reason for your answer to part I.

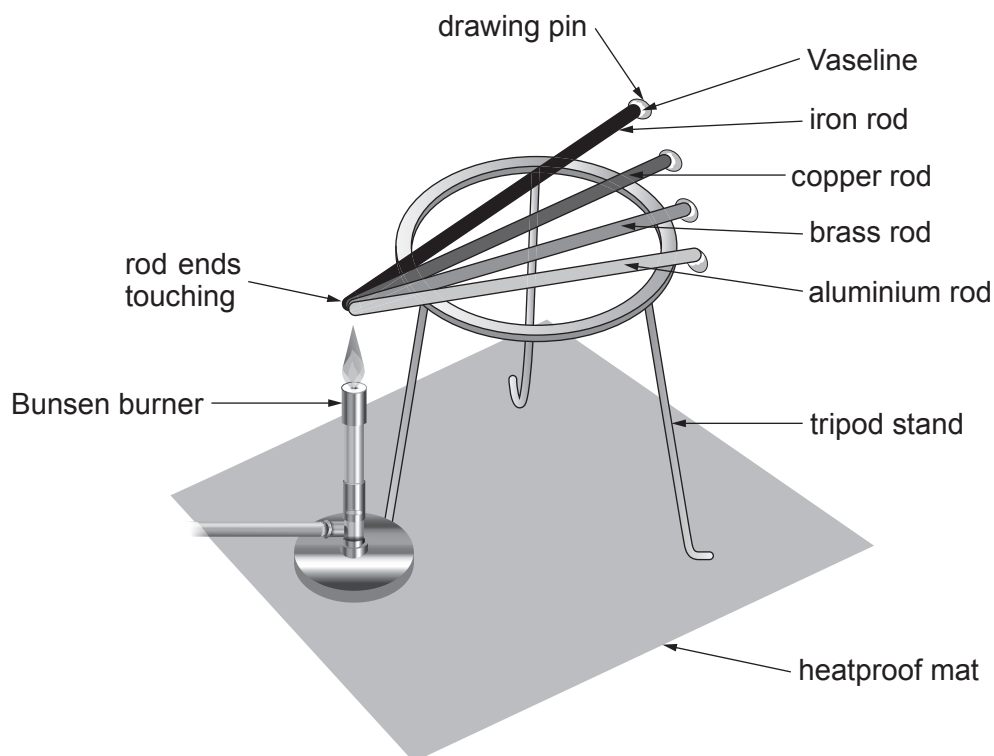
[1]

.....

.....



- (b) In their next experiment the students placed 4 metal rods on a tripod. The rods were identical except they were made from different metals. They attached a drawing pin to the end of each rod with Vaseline. They heated the other end of each of the rods and recorded the time for each pin to drop off.



The students recorded the following results:

Metal	Time for pin to drop off (s)
iron	46
copper	10
brass	23
aluminium	18

List the metals in order from best conductor to worst conductor.

[2]

Best conductor

.....

.....

Worst conductor

